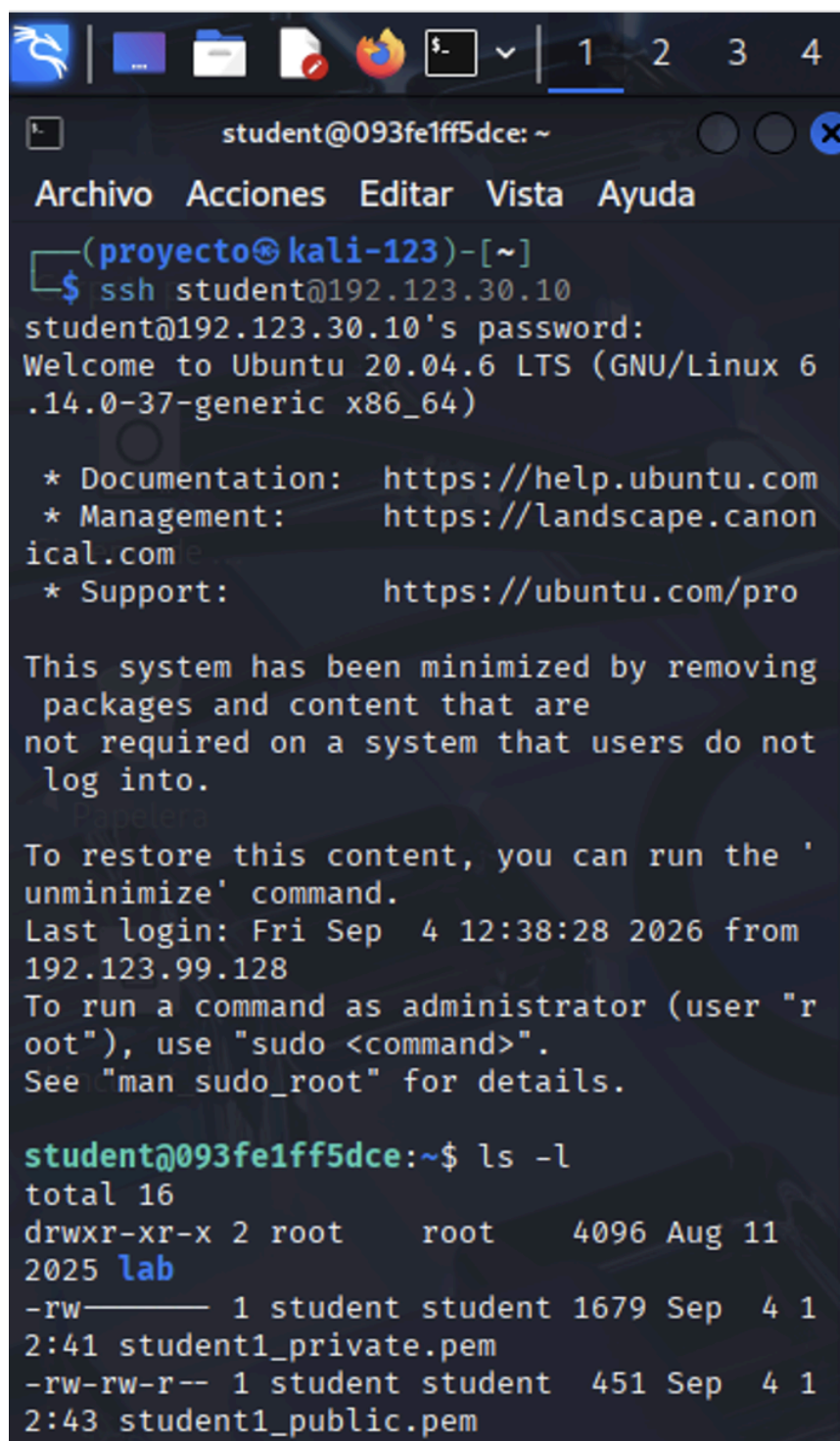


```
student@093fe1ff5dce: ~  
Archivo Acciones Editar Vista Ayuda  
(proyecto@kali-123)-[~]  
$ ssh student@192.123.30.10  
The authenticity of host '192.123.30.10 (192.123.30.10)' can't be established.  
ED25519 key fingerprint is SHA256:e72L4wbNnHn5zKMMFFBDxUY7fKkOVuifILsObfel800.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.123.30.10' (ED25519) to the list of known hosts.  
student@192.123.30.10's password:  
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 6.14.0-37-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
This system has been minimized by removing packages and content that are  
not required on a system that users do not log into.  
  
To restore this content, you can run the 'unminimize' command.  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
student@093fe1ff5dce:~$
```

```
student@79a9c1ef3ada: ~  
Archivo Acciones Editar Vista Ayuda  
(proyecto@kali-123)-[~]  
$ ssh student@192.123.30.11  
The authenticity of host '192.123.30.11 (192.123.30.11)' can't be established.  
ED25519 key fingerprint is SHA256:e72L4wbNnHn5zKMMFFBDxUY7fKkOVuifILsObfel800.  
This host key is known by the following other names/addresses:  
  ~/.ssh/known_hosts:18: [hashed name]  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.123.30.11' (ED25519) to the list of known hosts.  
student@192.123.30.11's password:  
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 6.14.0-37-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
This system has been minimized by removing packages and content that are  
not required on a system that users do not log into.  
  
To restore this content, you can run the 'unminimize' command.  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
student@79a9c1ef3ada:~$
```

10.10.20.63



```
student@093fe1ff5dce: ~  
Archivo Acciones Editar Vista Ayuda  
(proyecto@kali-123)-[~]  
$ ssh student@192.123.30.10  
student@192.123.30.10's password:  
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 6  
.14.0-37-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canon  
ical.com  
* Support:        https://ubuntu.com/pro  
  
This system has been minimized by removing  
packages and content that are  
not required on a system that users do not  
log into.  
  
To restore this content, you can run the '  
unminimize' command.  
Last login: Fri Sep  4 12:38:28 2026 from  
192.123.99.128  
To run a command as administrator (user "r  
oot"), use "sudo <command>".  
See "man sudo_root" for details.  
  
student@093fe1ff5dce:~$ ls -l  
total 16  
drwxr-xr-x 2 root    root    4096 Aug 11  
2025 lab  
-rw----- 1 student student 1679 Sep  4 1  
2:41 student1_private.pem  
-rw-rw-r-- 1 student student  451 Sep  4 1  
2:43 student1_public.pem
```



student@79a9c1ef3ada: ~



Archivo Acciones Editar Vista Ayuda

ED25519 key fingerprint is SHA256:e72L4wbN
This host key is known by the following ot
~/.ssh/known_hosts:18: [hashed name]
Are you sure you want to continue connecti
Warning: Permanently added '192.123.30.11'
student@192.123.30.11's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 6

* Documentation: <https://help.ubuntu.com>
* Management: <https://landscape.canon>
* Support: <https://ubuntu.com/pro>

This system has been minimized by removing
not required on a system that users do not

To restore this content, you can run the '

The programs included with the Ubuntu syst
the exact distribution terms for each prog
individual files in /usr/share/doc/*/copyr

Ubuntu comes with ABSOLUTELY NO WARRANTY,
applicable law.

To run a command as administrator (user "r
See "man sudo_root" for details.

```
student@79a9c1ef3ada:~$ openssl genrsa -ou  
t student2_private.pem 2048  
Generating RSA private key, 2048 bit long  
modulus (2 primes)
```

```
.....+++++  
.....+++++  
e is 65537 (0x010001)
```

```
student@79a9c1ef3ada:~$ openssl rsa -in st  
udent2_private.pem -pubout -out student2_p  
ublic.pem  
writing RSA key
```

```
student@79a9c1ef3ada:~$ ls -l student2_pri  
vate.pem student2_public.pem  
-rw----- 1 student student 1679 Sep  4 1  
2:46 student2_private.pem  
-rw-rw-r-- 1 student student  451 Sep  4 1  
2:47 student2_public.pem
```

```
student@79a9c1ef3ada:~$
```

```
10.10.20.63
/home/proyecto
Archivo Acciones Editar Vista Ayuda
This system has been minimized by removing packages and content that are not required on a system that users do not log into.
To restore this content, you can run the 'unminimize' command.
Last login: Fri Sep  4 12:38:28 2026 from 192.123.99.128
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

student@093fe1ff5dce:~$ ls -l
total 16
drwxr-xr-x 2 root    root    4096 Aug 11 2025 lab
-rw-r--r-- 1 student student 1679 Sep  4 12:41 student1_private.pem
-rw-rw-r-- 1 student student  451 Sep  4 12:43 student1_public.pem
-rw-rw-r-- 1 student student  451 Sep  4 12:42 student1_public.perm
student@093fe1ff5dce:~$ scp student1_public.pem student@192.123.30.11^C
student@093fe1ff5dce:~$ scp student1_public.pem student@192.123.30.11:~
The authenticity of host '192.123.30.11 (192.123.30.11)' can't be established.
ECDSA key fingerprint is SHA256:DEnmlzXVl8 1iZgewN2qAY3CfuNYiMcnkK0mYE1hZYPM.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.123.30.11' (ECDSA) to the list of known hosts.
student@192.123.30.11's password:
student 100% 451 893.3KB/s 00:00
student@093fe1ff5dce:~$ ls -l student2_public.pem
-rw-rw-r-- 1 student student 451 Sep  4 12:51 student2_public.pem
student@093fe1ff5dce:~$ ^C
student@093fe1ff5dce:~$ ^C

student@79a9c1ef3ada: ~
Archivo Acciones Editar Vista Ayuda
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

student@79a9c1ef3ada:~$ openssl genrsa -out student2_private.pem 2048
Generating RSA private key, 2048 bit long modulus (2 primes)
.....+++++
.....+++++
e is 65537 (0x010001)
student@79a9c1ef3ada:~$ openssl rsa -in student2_private.pem -pubout -out student2_public.pem
writing RSA key
student@79a9c1ef3ada:~$ ls -l student2_private.pem student2_public.pem
-rw-r--r-- 1 student student 1679 Sep  4 12:46 student2_private.pem
-rw-rw-r-- 1 student student  451 Sep  4 12:47 student2_public.pem
student@79a9c1ef3ada:~$ scp student2_public.pem student@192.123.30.10:~
The authenticity of host '192.123.30.10 (192.123.30.10)' can't be established.
ECDSA key fingerprint is SHA256:DEnmlzXVl8 1iZgewN2qAY3CfuNYiMcnkK0mYE1hZYPM.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.123.30.10' (ECDSA) to the list of known hosts.
student@192.123.30.10's password:
student 100% 451 952.6KB/s 00:00
student@79a9c1ef3ada:~$ ls -l student2_public.pem
-rw-rw-r-- 1 student student 451 Sep  4 12:47 student2_public.pem
student@79a9c1ef3ada:~$ ls -l student2_public.pem
-rw-rw-r-- 1 student student 451 Sep  4 12:47 student2_public.pem
student@79a9c1ef3ada:~$ ls -l student1_public.pem
-rw-rw-r-- 1 student student 451 Sep  4 12:50 student1_public.pem
```

```
sudo apt update && sudo apt install python3-pip -y
pip3 install pycryptodome --break-system-packages
#!/usr/bin/env python3
"""
```

sobre_digital.py

Lab 3 - Cifrado Asimetrico y Distribucion de Claves

Genera una clave simetrica AES aleatoria y la cifra con RSA usando la clave publica del destinatario. Tambien permite descifrarla con la clave privada del destinatario.

Uso:

En el REMITENTE (genera una clave AES nueva y la cifra con RSA)

```
python3 sobre_digital.py generar_y_cifrar <clave_publica_destinatario.pem>  
<clave_aes_cifrada.bin> <clave_aes_texto_plano.key>
```

```
# En el DESTINATARIO (descifra la clave AES con su clave privada)  
python3 sobre_digital.py descifrar_clave <clave_privada_propia.pem>  
<clave_aes_cifrada.bin> <clave_aes_recuperada.key>  
"""
```

```
import sys  
import os  
from Crypto.PublicKey import RSA  
from Crypto.Cipher import PKCS1_OAEP  
from Crypto.Random import get_random_bytes
```

```
def generar_y_cifrar(ruta_pub_destinatario: str, ruta_salida_cifrada: str,  
ruta_salida_clave_plana: str):  
    # 1. Generar una clave AES-256 aleatoria (32 bytes)  
    clave_aes = get_random_bytes(32)  
  
    # Guardamos la clave AES en texto plano (LOCAL, para usarla con cifrado_ftp.py)  
    # En un caso real, este archivo se borraría después de usarlo.  
    with open(ruta_salida_clave_plana, "wb") as f:  
        f.write(clave_aes)  
    print(f"[OK] Clave AES generada y guardada temporalmente en:  
{ruta_salida_clave_plana}")  
    print(f"    (Esta clave en texto plano es SOLO para uso local del remitente)")  
  
    # 2. Cargar la clave pública del destinatario  
    with open(ruta_pub_destinatario, "rb") as f:  
        clave_publica = RSA.import_key(f.read())  
  
    # 3. Cifrar la clave AES usando RSA (PKCS1_OAEP es el esquema recomendado)  
    cifrador_rsa = PKCS1_OAEP.new(clave_publica)  
    clave_aes_cifrada = cifrador_rsa.encrypt(clave_aes)  
  
    with open(ruta_salida_cifrada, "wb") as f:  
        f.write(clave_aes_cifrada)  
  
    print(f"[OK] Clave AES cifrada con RSA guardada en: {ruta_salida_cifrada}")  
    print(f"    Tamaño de la clave AES cifrada: {len(clave_aes_cifrada)} bytes")
```

```
def descifrar_clave(ruta_priv_propia: str, ruta_entrada_cifrada: str, ruta_salida_clave: str):  
    # 1. Cargar la clave privada propia  
    with open(ruta_priv_propia, "rb") as f:  
        clave_privada = RSA.import_key(f.read())
```

```

# 2. Leer la clave AES cifrada
with open(ruta_entrada_cifrada, "rb") as f:
    clave_aes_cifrada = f.read()

# 3. Descifrar usando RSA + la clave privada
descifrador_rsa = PKCS1_OAEP.new(clave_privada)
clave_aes = descifrador_rsa.decrypt(clave_aes_cifrada)

with open(ruta_salida_clave, "wb") as f:
    f.write(clave_aes)

print(f"[OK] Clave AES descifrada y guardada en: {ruta_salida_clave}")

```

```

def main():
    if len(sys.argv) != 5:
        print(__doc__)
        sys.exit(1)

    modo = sys.argv[1]

    if modo == "generar_y_cifrar":
        _, _, ruta_pub, ruta_salida_cifrada, ruta_salida_clave_plana = sys.argv
        if not os.path.isfile(ruta_pub):
            print(f"[ERROR] No existe: {ruta_pub}")
            sys.exit(1)
        generar_y_cifrar(ruta_pub, ruta_salida_cifrada, ruta_salida_clave_plana)

    elif modo == "descifrar_clave":
        _, _, ruta_priv, ruta_entrada_cifrada, ruta_salida_clave = sys.argv
        if not os.path.isfile(ruta_priv) or not os.path.isfile(ruta_entrada_cifrada):
            print("[ERROR] Alguno de los archivos de entrada no existe.")
            sys.exit(1)
        descifrar_clave(ruta_priv, ruta_entrada_cifrada, ruta_salida_clave)

    else:
        print("[ERROR] Modo no reconocido. Use 'generar_y_cifrar' o 'descifrar_clave'.")
        sys.exit(1)

if __name__ == "__main__":
    main()

```

```
student@093fe1ff5dce: ~
Archivo Acciones Editar Vista Ayuda

1.gz (of link group pico) doesn't exist
student@093fe1ff5dce:~$ nano cifrado_ftp.p
y
student@093fe1ff5dce:~$ echo "Mensaje secr
eto de student1 para student2" > mensaje.t
xt
student@093fe1ff5dce:~$ cat > sobre_digita
l.py << 'EOF'
> #!/usr/bin/env python3
> """
> sobre_digital.py
> Lab 3 - Cifrado Asimetrico y Distribucio
n de Claves
> Genera una clave simetrica AES aleatoria
y la cifra con RSA usando la
> clave publica del destinatario. Tambien
permite descifrarla con la
> clave privada del destinatario.
>
> Uso:
> # En el REMITENTE (genera una clave
AES nueva y la cifra con RSA)
> python3 sobre_digital.py generar_y_c
ifrar <clave_publica_destinatario.pem> <cl
ave_aes_cifrada.bin> <clave_aes_texto_plan
o.key>
>
> # En el DESTINATARIO (descifra la cl
ave AES con su clave privada)
> python3 sobre_digital.py descifrar_c
lave <clave_privada_propia.pem> <clave_aes
_cifrada.bin> <clave_aes_recuperada.key>
> """
>
> import sys
> import os
> from Crypto.PublicKey import RSA
> from Crypto.Cipher import PKCS1_OAEP
> from Crypto.Random import get_random_byt
es
>
>
> def generar_y_cifrar(ruta_pub_destinatar
io: str, ruta_salida_cifrada: str, ruta_sa
```

```
if __name__ == '__main__':  
    main()
```

```
student@093fe1ff5dce:~$ python3 sobre_digital.py generar_y_cifrar student2_public.pem clave_aes_cifrada.bin clave_aes.key
```

```
[OK] Clave AES generada y guardada temporalmente en: clave_aes.key
```

```
      (Esta clave en texto plano es SOLO para uso local del remitente)
```

```
[OK] Clave AES cifrada con RSA guardada en: clave_aes_cifrada.bin
```

```
      Tamano de la clave AES cifrada: 256 bytes
```

< student@093fe1ff5dce:~\$

Archivo Acciones Editar Vista Ayuda

```
> print(f"[OK] Archivo descifrado guardado en: {ruta_salida}")
>
>
> def main():
>     if len(sys.argv) != 4:
>         print(__doc__)
>         sys.exit(1)
>
>     modo, entrada, salida = sys.argv[1], sys.argv[2], sys.argv[3]
>
>     if not os.path.isfile(entrada):
>         print(f"[ERROR] El archivo de entrada no existe: {entrada}")
>         sys.exit(1)
>
>     password = getpass.getpass("Ingrese la contraseña de cifrado: ")
>
>     if modo == "cifrar":
>         cifrar_archivo(entrada, salida, password)
>     elif modo == "descifrar":
>         descifrar_archivo(entrada, salida, password)
>     else:
>         print("[ERROR] Modo no reconocido. Use 'cifrar' o 'descifrar'.")
>         sys.exit(1)
>
>
> if __name__ == "__main__":
>     main()
> EOF
student@093fe1ff5dce:~$ python3 -m py_compile cifrado_ftp.py
student@093fe1ff5dce:~$ echo "Mensaje secreto de student1 para student2" > mensaje.txt
student@093fe1ff5dce:~$ base64 clave_aes.key
vmvtW2r85RQgvDmWr6t5G0hYaqE0Ie8NsoELKUIg0lS=
```

```
student@093fe1ff5dce: ~  
Archivo Acciones Editar Vista Ayuda  
sys.argv[2], sys.argv[3]  
> if not os.path.isfile(entrada):  
>     print(f"[ERROR] El archivo de entrada no existe: {entrada}")  
>     sys.exit(1)  
>  
>     password = getpass.getpass("Ingrese la contraseña de cifrado: ")  
>  
>     if modo == "cifrar":  
>         cifrar_archivo(entrada, salida, password)  
>     elif modo == "descifrar":  
>         descifrar_archivo(entrada, salida, password)  
>     else:  
>         print("[ERROR] Modo no reconocido. Use 'cifrar' o 'descifrar'.")  
>         sys.exit(1)  
>  
>  
> if __name__ == "__main__":  
>     main()  
> EOF  
student@093fe1ff5dce:~$ python3 -m py_compile cifrado_ft.py  
student@093fe1ff5dce:~$ echo "Mensaje secreto de student1 para student2" > mensaje.txt  
student@093fe1ff5dce:~$ base64 clave_aes.key  
vmvtW2r85RQgvDmWr6t5G0hYaqE0Ie8NsoELKUIg0ls=  
student@093fe1ff5dce:~$ python3 cifrado_ft.py cifrar mensaje.txt mensaje.txt.enc  
Ingrese la contraseña de cifrado:  
[OK] Archivo cifrado guardado en: mensaje.txt.enc  
student@093fe1ff5dce:~$ cat mensaje.txt.enc  
♦♦PŘ♦  
S♦♦~♦♦`♦E♦8♦♦"T#♦uL♦♦♦♦H♦♦F; ♦♦♦Ku♦♦♦y♦T♦♦  
gk=,♦rqd"♦♦♦♦5♦♦♦k}i-0♦student@093fe1ff5dce:~$
```